



Original Article

Female countertactics to male feticide and infanticide in a multilevel primate society

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The occurrence of male feticide and/or infanticide represents an extreme case of sexual conflict: an adaptive strategy of male reproduction at the expense of females. Females are predicted to develop numerous countertactics; however, it has remained unclear whether countertactics can effectively shift the balance which likely depends on the social and mating system. We conducted a 15-year, detailed observation of female and male behavior during male replacement in free-ranging golden snub-nosed monkeys (*Rhinopithecus roxellana*) of a multilevel society. We have identified several potential countertactics against feticide and/or infanticide, including pregnant or lactating females staying with ousted males or transferring to stay with the ousted male after being attacked, which lead to 2 of 11 fetuses succumbing to feticide, and extra-unit mating, female joint defense and transferring with their infants to other social units, which lead to 6 of 34 infants succumbing to infanticide. Female joint defenses crucially delay infanticide, providing mothers the opportunity to transfer with their vulnerable infants to another social unit where they have previously mated with the male leader or where they have female relatives. Importantly, in response to mothers starting to transfer, some males cease attacking infants, a change in behavior that is likely to be beneficial to males because during their tenures they can sire future offspring of the mothers who remain. Our findings show how dynamic interactions between males and females can be and how the conditions in multilevel societies can shift the outcome of intersexual conflict from males to females.

Key words: female countertactics, male feticide, male infanticide, multilevel society, *Rhinopithecus roxellana*, sexual conflict.

INTRODUCTION

In sexually reproducing species, the reproductive strategies of females and males often diverge, creating conflict between the sexes (Parker 2006). Such conflict can result in antagonistic coevolution (Chapman and Partridge 1996; Rice 1996), where individuals of each sex are selected to maximize their fitness at the expense of their mates (Trivers 1972). Intersexual conflict has frequently been studied from the perspective of males to understand the conditions under which strategies involving harm to females are adaptive (Chapman et al. 2003) or in some case from the perspective of females to determine the conditions leading to

either successful resistance or to females gaining at the expense of males (Ebensperger 1998; Palombit 2015). However, most studies have focused on a single sex, limiting our understanding of interactions and potential dynamic switch in the use of tactics and countertactics.

There are many forms of sexual conflict in non-human primates, with sexually selected infanticide of unweaned infants by nonsire males the most ubiquitous example (Hrdy 1997; Palombit 2012; Lukas and Huchard 2014): killing unrelated infants shortens the inter-birth interval of the mother, permitting the killer to increase his reproductive success by siring her next infant sooner. In some species with frequent rank or resident status change among the males up to 50% of unweaned infants might be killed by males (Yin et al. 2013; Ma et al. 2019), generating considerable costs to females. However, another kind of sexual conflict similar to infanticide, sexually selected feticide, defined as “when males harass pregnant females with threats and aggression to the extent that females terminate pregnancies”

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(Zipple et al. 2019), has received much less attention. It has only been shown to occur in three species (Pereira 1983; Agoramoorthy et al. 1988; Alberts et al. 1992; Zipple et al. 2017; Bailey et al. 2021).

In turn, female primates exhibit a range of behavior that have been shown or suggested to be countertactics to avoid fetus or infant loss from male feticide or infanticide (Gross 1996). If females are pregnant during situations when infanticide is likely to occur, such as take-overs or male rank changes, another related phenomenon of male-mediated prenatal loss, the Bruce effect, defined as “when females terminate pregnancies after some form of sensory exposure (olfactory, visual, auditory, or tactile) to nonsire males” (Zipple et al. 2019), has been considered as a countertactic to avoid future infanticide by the male (Roberts et al. 2012). Although nonsire males may exhibit aggression toward females, aggression from males is not necessary to elicit the Bruce effect (Zipple et al. 2019) as it is spontaneous pregnancy loss (Roberts et al. 2012). Females might reduce the overall risk of infanticide by mating with many males, such that future dominants cannot be certain that a given offspring is not their own (Wolff and MacDonald 2004). If females have dependent offspring during these situations of male replacements or rank change of dominant males, they sometimes defend their offspring by separating it spatially from the infanticidal male, by counter-attacking infanticidal males on their own or in coalition with other females or a protector male, or by leaving the social unit and the infanticidal male, transferring with the ousted male, or joining a new social unit (Palombit 2012; Roberts et al. 2012; Lowe et al. 2019). Even though female countertactics have been reported in various species, few studies investigate their effectiveness and if and how males might react to these countertactics.

This balance in the sexual conflict likely depends on the social and mating system (Lukas and Huchard 2014). For example, in gregarious species, females might benefit from having improved detection and coalitionary defense against males that might be infanticidal (Nunn and van Schaik 2000; Zipple et al. 2017), but in itself, female coalitionary defense had been considered as ineffective because it can only delay infanticide (Palombit 2012). More effective tactics might be female promiscuity, for which females would need to have access to alternative mating partners in order to confuse paternity; protection from other males, which might be also limited in unimale societies; or be tolerated by other females in order to transfer into a new group/unit, which might be limited in territorial species. Female countertactics might be expected to be more effective if, in turn, males might gain benefits from stopping to attack infants. For example, if females would not leave a male who stopped attacking infants, this male might gain future reproductive opportunities when he can retain his tenure and mating control during future reproductions of these females. Accordingly, we would expect sexual conflict between males and females to be particularly dynamic in flexible, multilevel societies, where some individuals might have the opportunities to change different aspects of their behavior depending on the behavior of the individuals of the opposite sex.

Golden snub-nosed monkeys (*Rhinopithecus roxellana*) are Asian colobines endemic to China living in a multilevel society, in which females reside in core social units consisting primarily of one male unit (OMU) with multiple females, and multiple such OMUs associate and forage together as one larger social band (Xiang et al. 2014; Grueter et al. 2020). Infanticide has been observed when males take over the dominant position in OMUs (Yao et al. 2016), but most unweaned infants present during the times of male replacement survive without being attacked by the new male. Recently, Qi et al. (2020) suggest that females mate multiply with extra-group males in order

to reduce the risk of infanticide if those males later take over the females' social group, but detailed interactions and potential dynamic switch between males and females, that is, female joint defense, mothers transfer with their infants and males' response to these countertactics, remain unclear. In addition, the spatial association among and distribution of kin across units also means that females potentially have low costs of dispersal facilitating transfer between units (Zhao et al. 2008; Qi et al. 2009), but remain among close social partners which might provide support against attacks. Therefore, the social flexibility in snub-nosed monkeys might offer an opportunity to investigate the role of social interactions on the expression of male feticide/infanticide and female countertactics.

In this study, we firstly present preliminary data on male-mediated prenatal loss and try to identify if it is feticide or the Bruce effect; then we investigate female countertactics to male infanticide based on a 15-year, detailed observations of female and male behavior during take-overs and of the fate of fetus or infants in a free-ranging population of *R. roxellana*. From the view of males, the functional outcome of the Bruce effect is the same as the feticide which benefits for males. However, it is quite different from females' view because the Bruce effect is spontaneous pregnancy loss when the females confront immigrant male (Bruce 1959) and has been considered as a cost-mitigating strategy to avoid future infanticide by the male (Zipple 2020); it provides relative benefits to female that pregnancy termination following male replacement have greater reproductive success than females that subsequently lose their offspring to infanticide because they return to reproductive cycle earlier (Roberts et al. 2012). Therefore, to distinguish the Bruce effect from feticide, we expect that 1) high frequency of pregnancy termination combination with low rates of male–female aggression (Zipple et al. 2019), or 2) inter-birth interval of pregnant females that terminate pregnancies (the Bruce effect) would be shorter than those pregnant females incur infanticide.

We then hypothesize that females' countertactics would be versatile and effective. Females would copulate with males other than the leader of their OMU, including with potential killers (unassociated adult males in the all-male unit, who would take over their OMU) or with potential protectors (resident males of other OMUs, who could offer protection for the mother and unweaned infant if they immigrate into these OMUs). When the unweaned infant suffers an unsuccessful attack from new male, females would leave their original OMUs to seek male protection, either staying with the ousted male or immigrating into a new OMU where they have previously mated with the male leader or where they have female relatives. Females are able to defend offspring through joint defense (but see the opposite result in red howlers [Crockett et al. 2000] and yellow baboons [Zipple et al. 2017]), and larger size of OMU (members except the infant) would lead less risk of being killed or attacked because female joint defense might be stronger and/or early detection and vigilance might also be more efficient in larger OMU.

We next investigate strategies of males in response to the females' countertactics and their benefits. After male replacement, new resident males have two different behaviors, either tolerance or aggression, towards the unweaned infant. Therefore, as resident males are more likely to sire the next offspring of females in their OMUs (Yao et al. 2016), we predict that 1) tolerant males would benefit from tolerating or refraining to attack after lactating female transferring with unweaned infant if tenures of tolerance male are longer than females' birth intervals, and/or 2) tolerant male might have longer tenures than those of aggressive males as females play an important role during male replacement (Fang et al. 2018) and

are also observed to help resident male fight off males from AMU (Xiang ZF, personal observation). Furthermore, we also predict that 3) older males are more likely to be aggressive because they need immediate reproductive success and cannot wait for future reproductive success.

METHODS

Study sites and subjects

We conducted the study at Dalongtan (31°29'N, 110°18'E; elevation: 2200 m) in Shennongjia National Park, central China. The study population is free-ranging, but partly provisioned. Reserve staff began provisioning the monkeys in 2006 for the purpose of ecotourism (Zhang et al. 2019; Yao et al. 2021). Food was provided two to three times per day and included lichen, pine seeds, apples, carrots, oranges, and peaches; the monkeys were provisioned at different feeding sites/platforms. When not provisioned, the monkeys ranged across an area with a radius of ~ 1 km. Provisioning likely affected the monkeys' social behavior and daily activities, especially early in the process. However, they soon became habituated to humans, and instead of fleeing, the monkeys started to ignore the presence of researchers and reserve staff most of the time. All adult members were individually identifiable based on a unique set of physical features, such as body size, pelage color, evidence of injuries or scars, the shape and size of a female's nipples, and the shapes of granulomatous flanges which are present of fleshy nodules on both sides of the upper lip in adult males (Yu et al. 2013). The size of the focal band has increased from ~30 individuals comprising three OMUs and an associated AMU (five adult individuals) in January 2006 to > 90 individuals in seven OMUs and one AMU (>10 adult males) in December 2020. Males of AMU peripherally range around the band and periodically invade an OMU, seeking the opportunity to the replacement of the resident leader or mating with resident females (Xiang et al. 2014).

Data collection

We collected data on a daily basis at distances 5–50 m from 0800 to 1800 in winter–spring or 0700 to 1900 in summer–autumn between January 2006 and April 2021 (Chen et al. 2021a, 2021b). All instances of births, abortions, copulations, and male takeovers were observed using all-occurrence sampling. However, once a male takeover event had taken place, male attacks on pregnant females, infants, or mothers with infants, as well as any abortions, infanticide events were observed in a focal animal sampling. This focal animal sampling extended to the end of the mating (potential conception) season (late November) of the year of the takeover (Yao et al. 2016). All events were recorded via continuous all-occurrence sampling once an event began (Xiang et al. 2019). Generally, after the replacement event, the ousted male would leave the original unit and join the AMU, or occasionally wander around original unit; therefore, the final replacement of the previous male was decided to have occurred when the previous male left the original unit and joined the AMU or when copulations between the new male and the females were observed without disturbance by the previous male. A successful infanticide attempt was defined to have occurred when the infant was killed by the male during the first set of attack, otherwise aggressive bouts towards the infant were scored as unsuccessful infanticide attempts.

For the paternity analysis, hairs with intact roots were collected from all infants born during the study period, their mothers, and from all reproductively mature adult males in the focal band (Guo et al. 2015). Hair samples of adult individuals were obtained by

plucking them directly by hand (with gloves) whereas samples of young infants—who were difficult to approach—were obtained by using a pole with glue on its end which was brought in contact with the back of the target individuals (Guo et al. 2015). Paternity exclusion was established using 16 different microsatellite loci (D13S321, D10S1432, D1S533, D14S306, D6S493, D7S817, D10S676, D1S1656, D1S1665, D7S1826, D7S2204, D6S1056, D2S442, D5S1457, D6S474, D10S611), for details on the DNA protocols see Xiang et al. (2014) and Yang et al. (2014). The reliability of our genotyping results from hair samples was confirmed by comparing allele patterns with matched blood samples of five captive individuals: the agreement was 99.1%. Paternity for the 129 offspring born into the band from 2006 to 2020 was determined with the software CERVUS3.0 (Kalinowski et al. 2007) with confidence levels of >80% (relaxed confidence) and >95% (strict confidence). These inferences matched exclusion standards, and in all cases in which paternity was assigned to an extra-unit male the male residing within the social unit during the conception was excluded at two or more loci.

Data analysis

Extra-unit copulations and extra-unit paternity were judged to have occurred if the mate or the father resided in a different social unit. A chi-square test was used to identify whether 1) any extra-unit copulations primarily occurred between females and established males residing in one-male units or males residing in the all-male unit, and 2) frequencies of extra-unit copulation of those female involved in male replacement are higher than those not involved. The two-sample independent *t*-test was used to establish whether there were significant differences 1) in size of OMU (members except the infant) between OMUs in which infants were attacked or killed by males versus those in which infants did not suffer attacks, or 2) between females' birth interval and tenures of both tolerance and aggressive male.

RESULTS

Male replacement, pregnant females, unweaned infants and their fortunes

We observed 19 male replacement events involving 17 OMUs and 16 adult males between January 2006 and March 2021 (Case #1 – #19, Table S1). Two events occurred when no females were pregnant or caring for unweaned offspring (#7 and #18).

Eleven pregnant females were present during male replacement events. Two pregnant females (18.2%) terminated their pregnancies (#1 and #3) after being attacked by new males. Another two pregnant females were also attacked by the new males with one pregnancy female gave birth (#11) and the other returned to stay with ousted male and gave birth and weaned her offspring (#17). Totally, four pregnant females (4/11, 36.4%) were attacked; no other pregnant females were observed to be attacked by the new males, and all gave birth to their offspring. Furthermore, there is no significant difference of interbirth interval between females that prenatal loss and those females whose infant were killed by male infanticide ($t = -0.979$, $df = 5$, $P > 0.05$). All these observations indicate that prenatal loss would be sexually selected feticide rather than the Bruce effect.

Thirty-four unweaned infants were present following male replacements. Two of them and their mothers left the unit with the ousted male and survived. Although more than half of all remaining unweaned infants (18/34) were attacked by new males,

only four infants (#2, #8, #11, and #19) were killed by the first set of attack, and two more infant was killed after repeated attacks (#8 and #19). Therefore, most unweaned infants (28/34) present during male replacements survived throughout the mating season either because their mothers stayed with the ousted male, transferred to another social unit after an unsuccessful attack, or were tolerated by the new male (Figure 1 and Table S1).

Female counter tactics against male feticide/infanticide

Stay with the old (ousted) male

One pregnant female (1/11, 9.1%) and four unweaned infants (4/34, 11.8%) were protected by old males (Figure 1 and Table S1). Two mothers and their unweaned infants initially stayed with the old (ousted) males after male replacement (#12 and Xf19 in #17). Two mothers and unweaned infants initially stayed with new male and returned to stay with old male after unsuccessful infanticide attempts (Tt19 in #17 and Hhe20 in #19). One pregnant female initially stayed with new male, and returned to stay with old male after an unsuccessful feticide attempt (Nn19 in #17).

Extra-unit mating

In total, 4697 copulation events were observed. Only 317 of these copulation events were extra-unit matings (6.7%), and these involved almost exclusively males in other one-male units (309 events) rather than males in the all-male unit (8 events) ($\chi^2 = 182.00$, $P < 0.001$). A total number of 129 paternities were identified with

eight cases of extra-unit paternity (6.20%), which involved resident males in other OMUs. All new males who took over OMUs came from AMUs, and accordingly were not the fathers of any offspring in the OMUs they entered.

Of the nine mothers with unweaned infants who left their original OMU and transferred to another OMU following male replacement, two mothers (Ln in Case #6, Yb in case #16) had previously been observed mating with the resident male of the destination OMU into which they transferred, but no extra-unit paternities were identified.

Joint defense with other females

In cases of male attacks on unweaned infants, most females of the same OMU (82.9%, 63/76) were observed to join with other females against the new male (Table S1). The lactating female whose infant was under attack was the initiator of 70.6% (12/17) of coalitions as she, after firstly seizing her infant, started shouting at the perpetrator. Then, other females of the same unit (and sometimes also juveniles) ran to the mother and subsequently started vocalizing loudly at the male. In the remaining instances the mother was still seizing her infant whereas other female already started shouting at the male before being joined by others (including the mother). In a few cases, non-lactating female collectively chased the perpetrator and keep it away. In response to females' facing and shouting at the male, he usually retreated quickly and stayed away from the females for some time. Generally, the lactating female would transfer with unweaned infant following an unsuccessful attack, and subsequently, the remaining females with unweaned infants in the same

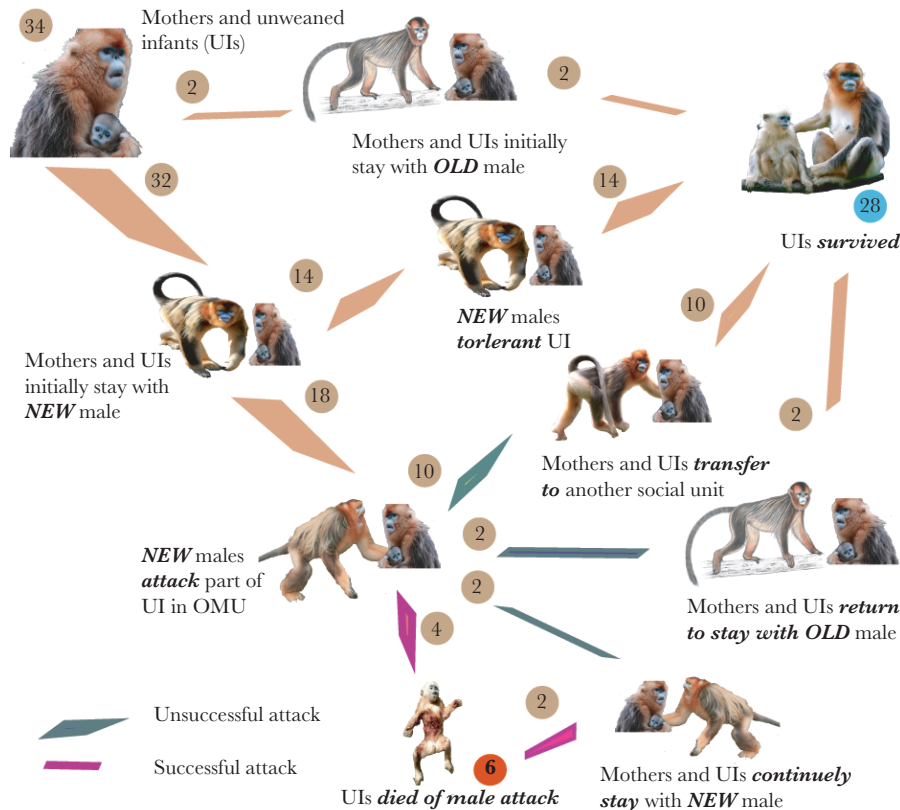


Figure 1

The fortune of unweaned infants (UIs) of *Rhinopithecus roxellana* following male takeovers: of 34 unweaned infants that were present during male replacement, 28 infants survived, and 4 infants died of male infanticide during the study period 2006 and 2021. (Number of females/infants observed showing a particular behavior are shown in circles).

OMU would maintain a relative longer distance from the new male. New males were not observed attacking any remaining unweaned infants.

The number of adult females involved in collective defense was 3.45 ± 0.91 (mean $\pm$ SD). The defensive behavior of females resulted in 81.8% (18/22) of the initial male attacks being unsuccessful, and since in the majority of cases mothers whose infants were attacked subsequently transferred to another OMU, most of these infants (12/18, 66.7%) survived to weaning. The effectiveness of female joint defense appears to depend on the size of OMU (the number of individuals except infants), as OMU size was smaller in OMUs in which attacks had been observed than in OMUs in which no attacks were observed (Figure 2a, $t = 3.95$, $df = 33$, $P < 0.001$), and the size of OMUs in which infants died from infanticide was also significantly smaller than those in which males never killed infants (Figure 2b, $t = 2.65$, $df = 29$, $P < 0.05$).

Female transfers with infants

Nearly a third of mothers (12/34) transferred to other OMUs or returned to stay with the old (ousted) male in response to the unsuccessful attacks on their offspring or another offspring in the OMU (Figure 1). The contexts of the ten mothers with dependent offspring who successfully immigrated into an already existing OMU included 1) six lactating females who had at least one close female relative (mother–daughter, sister–sister) in the new unit they joined, 2) two lactating females who had been observed mating with the resident male of destination unit, and 3) in two cases, one lactating female and her infant together with other females in the same OMU collectively joined another OMU as a clique.

Male strategies towards pregnant female/unweaned infant and its benefit

After male replacement, new males showed two strategies, either aggression or tolerance towards pregnant females/unweaned infants, which would bring different reproductive success. When males adopted a strategy of aggression, they obtained increased reproductive success through feticide or killing infants because

these males stayed in the group long enough to father the next offspring of victims' mother sooner (feticide: Case #1, #3; infanticide: Case #2, #8, #11, and #19). However, they lost benefits as the pregnant female transfer to stay with ousted male when attack did not result pregnancy termination (Case #17) or lactating females emigrated when the first set of attack did not kill the offspring (#4, #6, #17, and #19).

When males adopted a strategy of tolerance towards unweaned infant (#10, #12, and #13), they needed to wait longer until siring the next offspring of these females, but likely gained increased future reproductive benefits with those females who did not emigrate (#5, #9, and #14–16). The tolerant male would obtain reproductive benefit because tenures of tolerant males are significantly longer 1) than the inter-birth intervals which takes a female to successfully wean her current offspring ($t = 3.12$, $df = 53$, $P < 0.05$, Figure 3), and 2) than those of aggressive males ($t = 2.73$, $df = 10$, $P < 0.05$, Figure 3). Additionally, although infanticide might be expected to be favored by older males, who might rely more on immediate fitness benefits, aggressive males are not older than tolerant males ($t = 1.07$, $df = 15$, $P = 0.303$).

DISCUSSION

The detailed observations of female responses during male-takeovers across 15 years have observed two cases of feticide and six infanticide events, and it also suggests that living in a multilevel society means that females can mate with extra-unit males, form joint defense with other females in the same OMU against male attacks, and are able to successfully transfer to other OMUs to remove their unweaned offspring from infanticidal males after the infant was attacked by new male. These countertactics appear effective, as in response most males tolerated the dependent infant and/or stopped attacking any further infants when some mothers left with infants, and thus only a small number of infants (6/34) and fetuses (2/11) present during take-overs were killed by new males. Males in turn gain from changing their own strategy, as tolerating

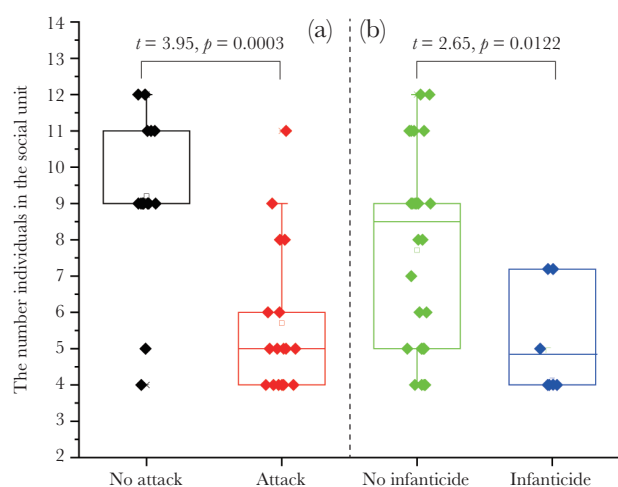


Figure 2

Differences in size of social unit (one male unit, OMU) with and without recorded of male attacks on infants and infanticide indicating an evidence for hypothesis that larger OMU size can lead to less risk of being killed (b) or attacked (a) because stronger female–female joint defense and vigilance of risk detection might also be more efficient.

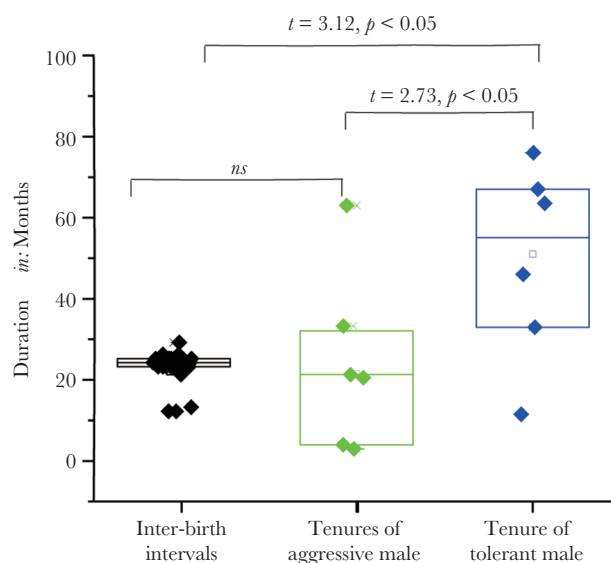


Figure 3

Difference of male tenures in different strategies to unweaned infants after male replacement indicating evidences for the hypothesis that tolerance males would benefit because their tenures (i) are longer than inter-birth intervals of female and (ii) longer than those of aggressive male.

unweaned infants means their mothers stay with them, and tolerant males would have a more chance of siring offspring with them in the future with longer tenure.

In response to the threat of infanticide by males, females have been argued to exhibit the Bruce effect, where they spontaneously terminate pregnancies after exposure to unfamiliar males (Bruce 1959) as an adaptive strategy as it is highly likely that a female's offspring will be killed after its birth (Roberts et al. 2012). This counter-strategy has been confirmed in other wild mammals including a wild primate, the gelada (*Theropithecus gelada*) with 80% of pregnancies terminated in the weeks after a dominant male is replaced. In other studies of Bruce effect (Berger 1983; Hackländer and Arnold 1999), more than 90% pregnant females induce abortions following male takeover. In the study population, the rate of fetal death (18.2%) is much lower than fetal death rates reported in those studies, which suggesting that fetal loss is not an obligate response to male takeovers. Therefore, prenatal loss in golden snub-nosed monkey might be the result of sexually selected feticide rather than the Bruce effect in instances where the new males are aggressive towards females and their offspring, as has been shown in Savannah baboon (Zipple et al. 2017).

In fact, the Bruce effect has likely failed to evolve due to several reasons in golden snub-nosed monkeys. First, there is no additional benefit for females to terminate pregnancy before August. Females could not be conceptive shortly after abortions because golden snub-nosed monkeys are strictly seasonal breeders with most conceptions occurring between late August and early November (Xiang et al. 2017). Our result indicates that there is no significant difference of interbirth interval between females that suffered prenatal loss and those females whose infant were killed by male infanticide. Second, the risk of male infanticide is relatively low because females have effective countertactics. Therefore, the best evolutionary strategy is not to terminate gestation because the pregnant females have many other countertactics to reduce the risk of male infanticide.

Zipple (2020) propose that male feticide might be selected over the Bruce effect for across a broader range of the mammalian taxonomy. However, despite that male infanticide has been reported in numerous species of primates (Palombit 2012, 2015), sexually selected feticide has received less attention due to the absence of a reliable pregnancy sign or demographic data for a wild population. Before this study, feticide only has been observed in primate population of Hanuman langurs *Presbytis entellus* (Agoramoorthy et al 1988), yellow baboons *Papio cynocephalus* (Pereira 1983, Alberts et al 1992, Zipple et al 2017) and olive baboons *Papio anubis* (Bailey et al 2021). Therefore, we hope more studies focus on prenatal loss in the future to facilitate our understanding evolutionary models of how sexual conflict and social systems interact.

Staying with ousted (old) male might be an effective tactic in species where there is high risk of infanticide after male replacement. In these circumstances, lactating female select to stay with ousted (old) male until her infant is weaned. This tactic had been observed in capuchin monkeys *Cebus capucinus* (Jack and Fedigan, 2009), red leaf monkeys *Presbytis rubicunda* (Davies 1987), purple-faced langurs *Presbytis senex* (Rudran 1973), Hanuman langur (Hrady 1997), and white-headed leaf monkeys *Trachypithecus leucocephalus* (Zhao et al. 2011). In this study, five cases had been observed (four mother-and-infant pairs and one pregnant female). In another study of same species, this behavior has been observed although the authors had not mentioned it is the tactic of infanticide avoidance (Fang et al. 2018).

Female primates appear best able to defend their offspring against male infanticide if they have the flexibility to interact with a range of social partners and rely on multiple countertactics. Multi-male mating by females in mammals has been indicated to be one of the most effective tactics to prevent infanticide, as it obscures paternity such that males refrain from killing offspring if they might be their father (Wolff and MacDonald 2004; Lukas and Huchard 2014). In snub-nosed monkeys, extra-OMU mating and extra-OMU paternities occur between females and males residing in other OMU (Guo et al. 2010; Yang et al. 2014; Qi et al. 2020; Xia et al. 2020). Furthermore, we had observed two cases where females were accepted into a unit with a resident male with whom she had previously been observed mating although there no paternities were identified; another case was also reported in a recent study (Qi et al. 2020). In contrast, females very rarely mated with males residing in the all-male unit, even though all males taking over OMUs came from this unit because resident males from multiple OMUs collectively show intense aggression towards these outside males (Xiang et al. 2014), presumably limiting opportunities for females to mate with them and to reduce the risk of infanticide potentially even further as it does in some polygynandrous societies.

Recently, a study of the same species had showed that 40.0% infants were sired by bachelor males in AMU and 17.1% infant were sired by resident males in neighbor OMUs (Qi et al. 2020). We propose that the difference of extrapair mating might be difference in mating monopoly of resident male due to group size rather than speculate that "the bachelor group being artificially reduced in number to avoid male conflict and injury" (Qi et al. 2020). The study population of Qi et al. (2020) includes at least 173 individuals (7–14 OMUs and 3–6 AMUs), which is much larger than that of our study (29–90 individual in 3–4 OMUs with only one AMU); this may increase the difficult for the resident males to exclude extrapair mating and it gets worse when there is a partially overlapping home range, seasonal fission–fusion events and the transfer of OMUs with another band includes 106 individuals (Qi et al. 2017). Furthermore, in our study population all resident males collectively act to keep AMU away to monopolize access to estrous females during the breeding season (Xiang et al. 2014) and this might have effectively reduced the opportunities of estrous females to engage in extrapair mating.

As there is frequently pronounced dimorphism between females and males in primates, it has been suggested that mothers would be unable to prevent infanticide physically (Ebensperger 1998; Palombit 2012). In this study, we find that, while females might be unable to resist a male attack on their own, whereas in nearly 82.9% of attacks females formed joint defenses that were effective in preventing males from killing their offspring in that instance. Therefore, although female joint defenses against male attacks have sometimes been considered ineffective because females might be unable to completely prevent males from killing offspring (de Waal 1997), our findings suggest that in snub-nosed monkeys it might create the crucial delay for females to exhibit other countertactics, in particular keeping their offspring alive until they might have the opportunities to transfer from their unit. This effect was more pronounced in larger units, potentially because more adult females in an OMU can lead to stronger female–female coalitions, vigilance and risk detection might also be more efficient (Palombit 2012). However, in ursine howler monkeys (*Alouatta arctoidea*), infanticide rates were significantly positively associated female group size (Crockett and Janson 2000), and similarly in *Presbytis entellus* with infanticide rates were lowest and in the smallest group and highest

in the largest group (Koenig and Borries 2001). Furthermore, both feticide and infanticide were more common when more pregnant and lactating females were present in yellow baboons (Zipple et al. 2017). There might be two reasons for the differences between these primates. Firstly, although more females (as a proxy of larger group size) in a uni-male or multimale society would facilitate greater benefits of predator protection, larger associations of females are more attractive to extra-group males for takeover in uni-male group or male rank variation in multimale group which result in subsequent infanticide. Secondly, these species are territorial species which might limit lactating females transfer with their infants as they are not tolerated by other females in a new group/unit.

By transferring from their social unit mothers can remove their unweaned offspring from a potential killer male (Hrdy 1997). Establishing a new unit with the ousted (old) male, as was observed in five cases here, is presumably more likely to occur in species in which the habitat is not saturated with groups that defend territories. Alternatively, females can attempt to transfer into an already established social unit where a male would be unlikely to kill the offspring (Steenbeek et al. 1999). However, in western gorillas (*Gorilla gorilla gorilla*) which lives in one-male groups, female secondary dispersal may not be effective countertactics to overcome the negative consequences of sexual conflicts (Manguette et al. 2019). In multilevel societies such as in golden snub-nosed monkeys, the potential costs of transferring into a new unit are likely to be reduced compared with other societies. Some females already disperse as sub-adults, and both natal and secondary dispersal of females mostly occurs into other social units within their natal band (Zhao et al. 2008, Guo et al. 2015). This means that females frequently have female relatives residing in neighbor OMUs, which might help them to join and be tolerated in these units. Some females have sexual experiences with other resident male in the band, this might also offer her and her unweaned infant a refuge when there is high risk of infanticide. Females transferred with their dependent infants had been observed in another study of *R. roxellana* (Zhao et al. 2008, Fang et al. 2018) although it had not been mentioned if infants were attacked by new male or with an aim of infanticide avoidance. In addition, food competition among the band member in this species is low because ubiquitous lichens and leaves form a major dietary constituent throughout the year (Li 2006), facilitating units to be in close proximity to potentially reduce predatory attacks through traveling, feeding, and resting within the same areas (Grueter et al. 2020).

The relative low costs associated with female transfers appear to influence male behavior, too. From the perspective of males taking over a social unit, emigration of adult females means that they would lose reproductive opportunities completely. While infanticide might increase current reproductive success if its mother remains in the OMU, it appears that on average males are likely to gain higher fitness if they refrain from attacking the unweaned infant. This is in the case of the study band when they do not kill their current offspring because tenures of tolerant male are significant longer than the female reproductive cycle, tolerant males even have longer tenure than those aggressive males, and males are likely to sire future offspring of females as the low probability of extra-unit paternity (Yang et al. 2014; Yao et al. 2016). Similarly, a resident male of an OMU might not only accept an immigrating lactating female because he previously mated with her, but also because the lactating female can increase his future reproductive success if he remains dominant for at least another reproductive cycle of that female. Therefore, there might be a trade-off when new males make

a decision to kill or not kill an unweaned infant as the unsuccessful attack would lead to female transfer. This may be best explanation for why most infants are tolerant by new males without any aggression and why most infants involved infanticide (4/6) are killed during the first attack.

Taken together, we show that females snub-nosed monkeys are able to mount effective countertactics against male infanticide including staying with ousted (old) male, extra-unit mating, joint defense and transfer, leading to mortality from infanticide for those infants involved male replacement (17.7%) in the study population being much lower than the mortality observed in some other primates, especially in some Asian colobines in which females live in single-male or multi-male societies. For example, in a uni-male and territorial society of white-headed langurs where countertactics of extra-group mating and females transfer with their dependent infants are limited, all eleven infants younger than six months died of possible infanticide after male takeovers (Yin et al. 2013) and six young infants (who are too young to abruptly wean) died of presumed infanticide when their mothers stayed with new male after male replacement (Zhao et al. 2011). In uni-male and multi-male society of hanuman langurs (Winkler et al. 1984; Sommer 1987; Borries 1997) where countertactics of females transfer with their dependent infants is limited, death rates of infant following male replacement were 37.9% (11/29), and 21.6% (8/37) which also higher than golden snub-nosed monkeys (17.6%, 6/34).

Furthermore, the mortality from infanticide for those infants involved male replacement in our study is much less in the continuum of a multilevel society. For example, in two well-detailed African papionins, gelada baboons *Theropithecus gelada* (Beehner and Bergman, 2008; Schneider-Crease et al. 2020) and hamadryas baboons *Papio hamadryas* (Swedell and Tesfaye, 2003; Swedell et al. 2011, 2014), 50% and 66.7% infants that involved male replacements were killed by new males. The difference of infanticide would be explained by the difference in female countertactics due to specific-species dispersal patterns and social bonding. In gelada baboons, female secondary dispersal is rare (le Roux et al. 2011) and extra-mating paternity is relatively low (Miller et al. 2021) due to female geladas in the OMU are closely relatives which facilitate to form coalitions against bachelor male harassment. In hamadryas baboons, females in same OMU are unrelated due to they are forced to disperse from their natal OMU by leader male and extra-mating paternity is low because no all-male group in this species (Swedell et al. 2011). Females transferred along with their dependent infants after male aggression had never been observed in two papionines although female coalitions have been observed (Plavcan et al. 1995). In contrast, evidences that a number of resident females (along with their dependent infants) migrate into another OMU or breeding band had been observed in this study and another provision and free ranging *R. roxellana* (Fang et al., 2018). Qi et al. (2020) declared that there is no behavioral or suspected evidence of infanticide because high paternity of extrapair mating. In fact, behavioral or suspected infanticide after male replacement had described in the study of Zhao et al. (2008). Therefore, we confirmed that it is flexible social connections in golden snub-nosed monkeys that facilitate females to conduct diversity countertactics (including extra-unit mating) against male infanticide.

Finally, our result highlights the importance of females having diverse social connections for shifting the balance in sexual conflict, indicating that not only the immediate but also the wider social environment can have important evolutionary consequences. Previous work has suggested that in multilevel societies males might gain

fitness benefits from cooperative defense against intruders or predators by tolerating a set of established neighbors (Xiang et al. 2014; Qi et al. 2017). Our findings suggest that females residing in these one-male units might similarly gain fitness benefits from tolerating the presence of other female groups within the same feeding area, as the non-aggressive relationships between females in different groups can lead to increased offspring survival.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

ETHICS APPROVAL

The authors declare that they have no competing interests. All research methods adhered to the Chinese legal requirements and were evaluated and approved by Central South University of Forestry & Technology's Institutional Animal Care and Use Committee (IACUC) (protocol #2012-018).

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Data accessibility: Analyses reported in this article can be reproduced using the data provided by Xiang et al. (2022).

AUTHOR CONTRIBUTIONS

ZX, YY and ML conceived the research design, analyzed the data, and wrote the paper jointly. ZX, HY, YY, QH and WY performed the field observation, and ZX and ML conducted the paternity analysis.

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